

Hey, everybody. Welcome to your Chapter 11 review. In this episode, we'll be looking at those all-important health, wellness, and fitness assessments and the relationship they have to your clients' goals and optimal health. We all agree structured exercise and physical activity can reduce the risk for mortality and morbidity, while also providing many physiological, emotional, and cognitive benefits. The benefits can come from light or moderate intensity, all the way up to the really vigorous stuff.

However, any intensity of exercise can hold some risk. So all new clients, especially those planning to participate in more moderate to vigorous intensities of exercise, should be appropriately screened to evaluate the benefits versus the risks of participation. These assessments are your baseline for what's to come, and can be used to determine where your client wants to go. If you know those two things, then you can design the most appropriate program for their needs, using your knowledge and understanding of exercise as the foundation.

Assessments may also serve other important functions, such as identifying individuals at increased risk. These individuals may need to first undergo a more thorough medical evaluation, or in some cases, be excluded from exercise until conditions are under control or corrected. Assessments can even serve as motivators through realistic goal setting.

It's a trainer's legal and ethical responsibility to conduct a health risk assessment prior to having a client initiate an exercise program. The timing, relevance, and appropriateness of any physiological assessment should be considered. So by the end of this chapter, you need to be able to explain the importance of fitness assessments surrounding clients' goals and optimal health; collect and summarize subjective information from clients, using health screening assessments; and identify the steps for conducting physiological, body composition, and cardiorespiratory assessments appropriate to specific client types and goals.

Everything we talk about and teach means nothing without a thorough, efficient, and thoughtful preliminary assessment. This consists of questionnaires to collect information regarding existing risk factors for participation in exercise and physical activity to determine the need for medical clearance beforehand. Screening procedures should be valid, simple, cost and time efficient, and appropriate for the target population.

You have to conduct a Health Risk Assessment, or HRA, prior to having anyone start an exercise program. But keep in mind that new exercisers may be apprehensive, which could increase their anxieties about moving forward. The PAR-Q+ is considered the appropriate minimal screening tool. Make sure you get familiar with it, since literally everyone you see will fill this out.

A more thorough pre-exercise Health History Questionnaire, or HHQ, can be used to complement the PAR-Q+. The Health History Questionnaire includes information about a client's health history, lifestyle habits, and exercise and medical history. Without this information, you might be missing really important pieces of the puzzle. And ultimately, this could fundamentally change your approach in working with that individual.

Once your clients are cleared, you may next choose to complete physiological assessments. Assessments chosen should always consider relevance, appropriateness, validity, reliability, and sequence to get your client to their best possible outcome. The goal of using fitness assessments is to establish baseline measurements, which influence optimal program design and provide a means to measure and evaluate progress.

First, you can monitor heart rate. Heart rate is a valid indicator of work intensity or stress, both at rest and during exercise. It can also provide valuable feedback about cardiovascular adaptations to exercise over time.

There are several locations on the body that can be used to measure a client's heart rate. For accuracy, safety, and to make things easier on everyone, NASM recommends fitness professionals measure the radial pulse at the wrist. Blood pressure, or BP, is another important cardiovascular measurement, which is defined as the outward pressure exerted by blood upon the arterial walls. BP scores are important because higher scores indicate greater risks for developing cardiovascular disease, which can become life threatening.

It's important to understand the different ranges for blood pressure so you can understand the factors for potential health risks. Remember, if anyone has any of these risk factors, it is your job as a CPT to refer them out to a licensed medical professional. It is recommended that all fitness professionals attend a live training class to learn how to properly assess blood pressure.

Anthropometry and body composition measurements are another important assessment category for fitness professionals to understand. Basically, anthropometry is taking measurements for the purposes of understanding physical variation in size, weight, and proportion. These types of assessments are often used to quantify body composition through techniques like body fat assessment, body mass index, also known as BMI, or circumference measurements. Each measurement can provide valuable estimates of mortality and morbidity.

Now, although BMI and circumference measurements provide estimates of body shape, they are actually not the best predictors of a person's body fat percentage. You have to take the individual into account with all of these strategies. Skinfold measurements and hydrostatic or underwater weighting are also considered valid estimators of body composition, but they require more skill, and you'll need expensive equipment. While all of these methods are valid and will be useful to you at some point in your career, bioelectrical impedance analysis, or BIA, is the one that has become the most popular method used in fitness facilities due to its ease of use.

Next up is cardio. Cardiorespiratory assessments are used to identify safe and effective starting exercise intensities, as well as appropriate modes of cardio exercise. They can use multiple formats, such as treadmills, step machines, or bicycles, and can range from simple to complex, based on the client's abilities and goals. Some examples of cardio assessments you should know include VO2-Max testing, the YMCA 3-Minute Step Test, the Rockport Walk Test, the 1.5-mile run test, and the talk test.

The VO2-Max test is the gold standard marker of cardio fitness. It's also a marker of health, mortality, and morbidity, and may be the single most effective predictor of heart function. It measures the body's utilization rate of oxygen and reflects the body's ability to deliver oxygen to the exercising muscles and use it to manufacture energy.

A subjective rating of perceived exertion, or RPE, is a technique used to express or validate how hard a client feels that he or she is working during exercise by subjectively rating the difficulty of the exercise as they perceive it. The YMCA 3-Minute Step Test assesses aerobic fitness of deconditioned clients by having them complete three minutes of continuous step activity to evaluate recovery heart rate. It's considered suitable for low risk, apparently healthy, non-athletic adults.

For new clients who may not be very fit, a walking test is considered an appropriate cardiorespiratory assessment. A Rockport Walk Test is a one-mile walking test that predicts maximal oxygen consumption from either a timed performance or a heart rate response. For more fit populations, the 1.5-mile run test can be used to measure aerobic endurance by either scoring the individual's timed performance or their heart rate response.

Because ventilation provides a good indication of metabolism, the talk test for VT1 has been validated as a good estimate of aerobic efficiency. It gradually progresses in intensity level and relies on the interpretation of how a person talks when the body's metabolism undergoes a significant change. VT2 is going to be the go to for your more performance-centric clients by measuring the level at which the body can work at its highest sustainable, steady-state intensity for more than a few minutes.

As always, if you uncover anything during your assessments that falls out of your scope of practice, refer your client to the appropriate Allied medical health professional. There are countless ways to get your clients to where they want to go, but you must first know where to start. Once you have an understanding of your client's fitness level and goals, possible gaps in health, diet or previous exercise, and their potential risks and limitations, you can take all of that information and couple it with your exercise knowledge to design the perfect program just for them.

Not only is it something you're legally and ethically bound to do, it's a matter of what we always come back to, which is doing right by your clients and helping them live better lives. And that's it for Chapter 11. Great job sticking with it. Be well, and happy studying.